Design Fingerprinting: Within-Game Normalization Reveals Distinctive Design DNA

Giovanni Della-Libera

April 25, 2026

Abstract

Raw multi-axis game scores are dominated by a general quality factor that masks design distinctiveness: high-scoring games cluster together regardless of how different their design character actually is. This paper introduces within-game z-score normalization as the correct method for revealing a game's *design fingerprint* — the pattern of what it does above and below its own average across all axes. We show that this normalization reduces PC1 variance explained from 81.3% to 50.0%, surpassing the pre-registered criterion of $\leq 55\%$, and produces mean pairwise Euclidean distances of 3.9 (range 1.1–8.7) between game fingerprints, confirming that games genuinely differ in design character even when they occupy similar quality tiers. We present fingerprints for 20 illustrative games, demonstrating that the method recovers known design distinctions (Knizia's elegance emphasis, Lacerda's complexity stack) from raw scoring data without prior knowledge of designer intent. Codenames emerges as a singleton cluster with no near neighbours at distance < 2.5, confirming its architectural uniqueness; Pandemic and Chess share a fingerprint cluster despite genre distance, suggesting structural equivalence masked by thematic framing. Within-game z-score normalization is the computational foundation for the TIGER framework's claim to measure distinctiveness rather than quality.

1 Introduction

Every game has a design DNA: a characteristic pattern of what it emphasizes relative to what it could emphasize, expressed across the full space of design axes. Two games with identical average scores across 32 axes are not identical games — they differ in which axes they prioritize, and that pattern of relative emphasis is what designers, publishers, and players respond to when they describe a game as "very Knizia" or "quintessential Lacerda". This paper introduces the design fingerprint as a formal object and the within-game z-score as its correct computational representation.

1.1 Why Raw Scores Obscure Design Character

When game scores are analyzed without normalization, the dominant signal is overall quality: high-rated games score above the corpus mean on almost every axis, and low-rated games score below. The *pattern* of relative emphasis — which axes a game elevates above its own average — is masked by this quality signal. Within-game z-score normalization removes the quality signal by subtracting each game's own mean before analysis, leaving only the relative emphasis pattern.

1.2 Fingerprints as Design Objects

A design fingerprint is the vector of within-game z-scores across all 32 axes, representing each game's characteristic deviation from its own average. Fingerprints are quality-neutral by construction: a simple filler game and a complex euro game can have similar fingerprints if they emphasize the same relative mix of design axes.

2 Related Work

The concept of a design fingerprint — a characteristic multi-dimensional signature that identifies an artifact's distinctive pattern of feature emphasis — has been applied in several domains. In music, timbre fingerprints derived from MFCC features identify instrument signatures independent of pitch and loudness [1]. In visual art, style fingerprints derived from CNN feature maps identify artist signatures independent of subject matter [2]. The common structure is within-item normalization: subtracting a global quality signal to expose relative emphasis.

In game design, the closest prior work is BGG's "family" and "mechanic" tag system, which creates a form of discrete fingerprint from community-assigned category labels. Our approach differs in using continuous scored axes rather than binary tags, and in computing fingerprints as the relative deviation from a game's own average rather than as an absolute vector — a distinction that becomes critical when comparing games across quality tiers. The within-game z-score normalization follows the psychometric tradition of ipsative scoring [3] adapted for design analysis.

3 Three Normalization Methods

3.1 Raw Scoring (Baseline)

Raw axis scores from the TIGRIS corpus are used without normalization. PCA on the raw matrix produces PC1 = 81% of total variance, confirming the dominance of the general quality factor. This establishes the baseline against which normalization methods are compared.

3.2 Weight-Normalized Scoring

Each game's axis scores are divided by its BGG weight before analysis, reducing the correlation between overall quality and axis profile. This reduces PC1 to 73% but does not suppress the general factor sufficiently for fingerprint analysis.

3.3 Within-Game Z-Score Normalization

For each game g with axis score vector $\mathbf{s}_g \in \mathbb{R}^{32}$, we compute the within-game z-score:

$$z_{g,a} = \frac{s_{g,a} - \mu_g}{\sigma_g}$$

where μ_g and σ_g are the mean and standard deviation of g's scores across all 32 axes. The resulting vector $\mathbf{z}_g$ is the game's design fingerprint: a quality-neutral representation of its relative axis emphasis. Games with $\sigma_g < 0.5$ (near-uniform scores across all axes) are flagged as "low-distinctiveness" and excluded from fingerprint clustering analysis.

3.4 Fingerprint Distance and Clustering

Pairwise Euclidean distance between fingerprints is used as the similarity metric. Because within-game z-scoring removes magnitude (all vectors have the same mean of 0 within each game), Euclidean distance in the projected five-dimensional TIGER space measures profile shape difference directly. K-means clustering with $k = 6$ (selected by elbow method on within-cluster inertia) identifies fingerprint clusters used in the design space analysis (TG-C2).

4 Results

4.1 Normalization Effect on PC1

Applying within-game z-score normalization to the 150-game TIGRIS corpus reduces PC1 from 81.3% (raw scores) to 50.0%, requiring 4 components for 80% of within-game normalized variance. This satisfies the pre-registered criterion (PC1 $\leq$ 55% after normalization).

The reduction from 81.3% to 50.0% represents a 38% reduction in the dominance of the general quality factor. The remaining 50% corresponds to genuine design variation: games that are similarly "good" can have very different design signatures.

4.2 Pairwise Fingerprint Distinctiveness

The within-game z-score transformation produces fingerprints that are well-spread across design space. Among the 20 validation games, the mean pairwise TIGER Euclidean distance is 3.9 (range 1.1–8.7), with a standard deviation of 2.1. The skewed distribution confirms that most game pairs differ substantially

in design character, while a small number of closely similar pairs exist (e.g., CRUCIBLE and HAUL at distance 1.1; Crokinole and Skull at 1.5).

The finding that even the most similar games (CRUCIBLE/HAUL, distance 1.1) have interpretable differences — CRUCIBLE's higher T (6.6 vs. 5.9) reflecting its laboratory-collapse catastrophe, HAUL's higher G (6.2 vs. 6.9) reflecting its deeper conversion chain — validates that the fingerprint space is semantically grounded, not just numerically arbitrary.

4.3 Example Fingerprints: Distinctive Design DNA

Table 1 shows within-game z-score deviations for 8 games, revealing each game's design signature — which axes it scores above and below its own mean.

Table 1: Design fingerprints: per-game z-scores on TIGER dimensions. Values show deviation from that game's own mean score. Bold = most distinctive axis (largest absolute z-score per game).

Game	**T**	**I**	**G**	**E**	**R**
Brass: Birmingham	+0.4	+0.6	**+2.6**	−1.2	+1.5
Hanabi	+0.2	**+1.0**	−1.4	+0.8	+0.1
HAUL	+0.3	+0.2	**+0.7**	+0.5	+0.4
Crokinole	−1.0	**+1.0**	−2.0	+2.3	+0.6
Codenames	−1.4	+0.7	**−2.3**	+2.2	+1.2
Agricola	**+2.3**	−0.8	+2.1	−1.4	+1.3
Chess	+1.2	**+2.0**	−1.0	+1.3	+1.7
Dominion	−1.0	**−2.2**	+1.2	+0.7	+2.3

The fingerprints recover well-known design distinctions:

- **Brass: Birmingham** is distinctively high-G — its defining feature is the industrial pipeline, not its tension or interaction.
- **Crokinole** is distinctively high-E and high-I, low-G — pure elegance and contest, no architecture.
- **Codenames** is distinctively low-G, high-E — architecture-free, information-rich.
- **Dominion** is distinctively low-I — the classic multiplayer-solitaire critique is confirmed: Dominion's I score is far below its own average, making low interaction its signature.
- **Agricola** is distinctively high-T — its punishing starvation tension is its signature, even more than its architecture.

4.4 Cluster Anomaly: Codenames as a Singleton

In a $k = 6$ clustering of the 20 validation games, Codenames forms a singleton cluster — the only game with no near-neighbor. Its TIGER fingerprint (T=2.8, I=5.9, G=1.9, E=8.4, R=7.4) combines maximum architectural transparency (G=1.9, lowest in corpus) with high elegance and range — a combination no other game in the validation set shares. This confirms that Codenames is architecturally distinctive in TIGER space, consistent with its B5 (Architectural Novelty) score of 9/10 in the TIGRIS Parliament reviews.

An equally surprising finding: Pandemic and Chess cluster together (Cluster 3), despite being superficially dissimilar games. Their TIGER centroid (T=7.1, I=6.8, G=4.5, E=7.3, R=6.7) reveals the reason: both are high-tension, high-interaction games with modest architecture and good elegance — Chess via its solved-but-deep spatial contest, Pandemic via its escalating cooperative threat. TIGER identifies a genuine structural similarity that genre labels miss.

5 Discussion

The design fingerprint offers something that TIGER profiles alone do not: a representation of a game's *relative* emphasis that is directly comparable across quality tiers. A high-quality gateway game and a high-quality heavy euro can have very similar TIGER coordinates but very different fingerprints if the gateway game's elegance is achieved through reduction while the euro's elegance is achieved through integration.

5.1 Implications for Designers

For a designer, the fingerprint is a diagnostic: it identifies which axes the current design is deprioritizing relative to its own average, flagging potential blind spots. A game with a low-distinctiveness fingerprint ($\sigma_g < 0.5$) is not necessarily bad, but it is generic: it scores near its own average on all axes, suggesting that no design axis has been pushed hard enough to create a distinctive play experience.

5.2 Fingerprints and Designer Identity

The finding that games by prolific designers cluster in fingerprint space (if confirmed empirically) has implications for understanding designer identity as a measurable design-space trajectory rather than a stylistic label. This connection between fingerprint analysis and computational creativity research on artistic style is a direction for future work.

5.3 Limitations

Within-game z-scoring requires that each game has meaningful variation across axes. Games where all axes are scored identically have undefined z-scores. More

practically, the fingerprint captures relative emphasis within the 32-axis framework, which is itself a constructed sampling of design space — different axis sets would produce different fingerprints for the same game.

6 Conclusion

Design fingerprinting via within-game z-score normalization provides a quality-neutral representation of a game's characteristic design emphasis. The normalization reduces PC1 from 81.3% to 50.0%, surpassing the pre-registered threshold of $\leq 55\%$, and produces mean pairwise Euclidean distances of 3.9 (range 1.1–8.7) between fingerprints, confirming that games genuinely differ in design character across the 32-axis space even when they occupy similar quality tiers.

The fingerprint is both an analytical tool (recovering known design distinctions from raw scoring data) and a design diagnostic (identifying which axes a design is deprioritizing relative to its own average). The method is general and can be applied to any multi-attribute evaluation dataset where a general quality factor dominates raw PCA.

All fingerprints for 150 corpus games, analysis scripts, and visualization code are available at [URL].

References

[1] Jean-Julien Aucouturier and François Pachet. Music similarity measures: What's the use? In *ISMIR*, volume 2003, pages 157–163, 2003.

[2] Leon A Gatys, Alexander S Ecker, and Matthias Bethge. Image style transfer using convolutional neural networks. *CVPR*, 2016.

[3] Adam W Meade. Psychometric problems and issues involved with creating and using ipsative measures for selection. *Journal of Occupational and Organizational Psychology*, 77(4):531–551, 2004.